Deep Learning for Emotion Detection

Real-Time Mental Health Monitoring for Astronauts Using Deep Convolutional Neural Networks (CNNs)

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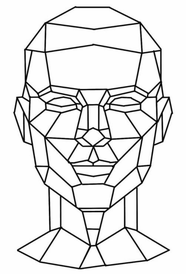


Figure 1 – Logo created by Finalist, 2025

BACKGROUND RESEARCH

Psychological & Physiological

Long-duration space missions can significantly impact astronauts' mental health due to isolation, confinement, disrupted sleep cycles, and exposure to microgravity. These factors can lead to emotional stress, cognitive challenges, sleep disturbances, and psychological conditions like anxiety and depression.

Common warning signs of mental health issues include:

- Changes in sleep or appetite
- Mood fluctuations
- Withdrawal or apathy
- Decline in cognitive function, memory, or concentration
- Increased sensitivity to stimuli
- Anxiety or unusual behaviors

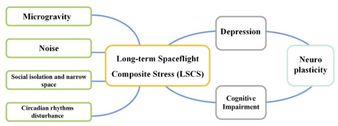


Figure 2 - Long-term Spaceflight Composite Stress [1]

NASA's Project Mercury (flights compared to simulation)

- Body temperature slightly increased
- Blood pressure increased
- Respiration was higher at lift-off

AI Models

A **FER Human-Robot** study used a two-stage CNN pipeline, a neural network that first detects & aligns faces which the second then uses to analyze emotions. The names of the two different neutral networks that were used were: ResNet Inception Based Faster R-CNN Model and Inception Based SSD Model.

- Both had about the same total loss and accuracy rate
- Accuracy rates were ~83%

Emotion Recognition

- Emotion detection using voice data and analyzing speech data
- Dataset included 40 commands in five emotions (angry, happy, sad, bored, and neutral) and had 14 speakers (7 male, 7 female)
- Achieved an 80% accuracy rate

Measuring Depression using Body Movements AI

- Study used 40-60 minute recordings of clinically diagnosed patients of Major Depressive Disorder (MDD)
- Model prediction and accuracy was compared to patients' self-reported scales
- Model was trained on 2013 FER dataset and resulted In a 70.71% accuracy rate

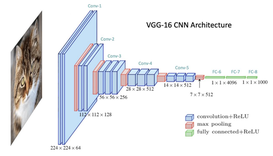


Figure 3 – Convolution Neural Networks from OpenCV [2]

RESEARCH OBJECTIVES

1. **PROBLEM STATEMENT –**
   - On long duration space missions, the psychological health of the astronauts is a critical issue. Assessing the mood and emotions of the crew member can improve astronaut performance and mission success.
2. **RESEARCH QUESTION –**
   - How can the use of AI track the psychological impacts of isolation, stressful missions, and physical implications of a microgravity environment through assessing astronauts' emotions using onboard software?
3. **ENGINEERING GOAL –**
   - Use generative AI to determine the current mood of the subject through building a facial AI model that assesses the mood and emotions of astronauts.
4. **ETHICAL CONSIDERATIONS –**
   - Dataset must include racial and gender diversity
   - Data must only be used for psychological monitoring and will not be shared with third parties
   - All data will be anonymized, no personally identifiable information will be linked to image data and labels
5. **EXPECTED OUTCOMES –**
   - A functional AI model capable of detecting emotions with high accuracy (ranging from 60% - 80%)
   - A real-time monitoring tool (i.e. a Streamlit app) that detects subject emotion and provides feedback for the model
6. **POTENTIAL CHALLENGES –**
   - Skew in the dataset could create bias
   - Running the AI model in real time may require significant computational resources, which may be limited on spacecrafts
   - Fluid shifts in microgravity may alter facial appearance, affecting the model's predictions
   - The model may overfit or underfit to the training data, performing well on the dataset but poorly in real-world scenarios
   - Image lighting and quality may impact the model's performance

ENGINEERING METHODOLOGY

Select a Dataset

Select a Programming Language. (Machine Learning models are usually programmed with Python or R.)

Research Datasets on:
- Kaggle
- Hugging Face
- Google Datasets
- Papers with Code
- Github

Datasets:
- Google API Open Datasets – Human Face Dataset
- Kaggle – A version of FER 2013 Dataset
- Kaggle – Mental Health Chatbot
- Chicago Face Database (CFD)

Programming Language: Python

Chosen dataset: Chicago Face Database (CFD)

Data Preprocessing

Visualize images from the dataset and plot images into a graph using PyPlot



Figure 4 & 5 – Visualized Images from CFD Dataset Created by Finalist, 2025

Resize the images to feed into the model (300, 200)

Each img had its label in the file name, so our team made empty lists to append the images to. Then they were enumerated

Build the Model + Train & Test Model

Create model architecture & check for trainable parameters



Figure 6 & 7 – Model Architecture & Parameter Calculation Created by Finalist, 2025

All parameters (features) from this dataset are trainable (6,688,132 out of 6,688,132 are trainable)

Split images into train and test variables



Figure 8 – Training & Test Variables Created by Finalist, 2025

Build App & Collect More Data

Create a Streamlit app powered by ngrok to demonstrate real-time emotion prediction using user-captured images. Correct predictions validate the model's accuracy; incorrect ones are added to a dataset predictions are used for retraining, improving performance over time through continuous learning.

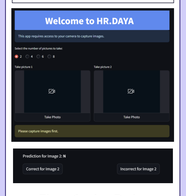


Figure 9 – App Design Created by Finalist, 2025

STATISTICAL ANALYSIS

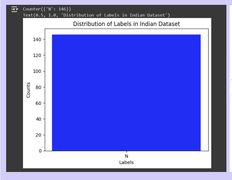


Figure 10 – Training & Test Variables, Created by Finalist, 2025

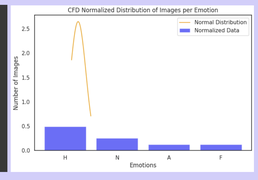


Figure 11 – Normal Distribution of Images per Emotion, Created by Finalist, 2025

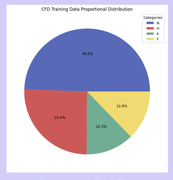


Figure 12 – Pie Chart Distribution of Imgs per Emotion, Created by Finalist, 2025

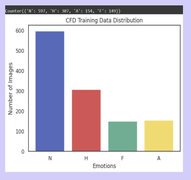


Figure 13 – Bar Chart Distribution of Images per Emotion, Created by Finalist, 2025

Figures 10 – 13 demonstrates the split in the training data images in the CFD dataset. Figure 10 shows the split in the Indian dataset, where all the image labels were neutral, which is why we chose not to utilize the Indian dataset. Figures 11-13 depict the split in the CFD dataset, our team decided to use through a normal distribution, a pie chart, and a bar graph. This

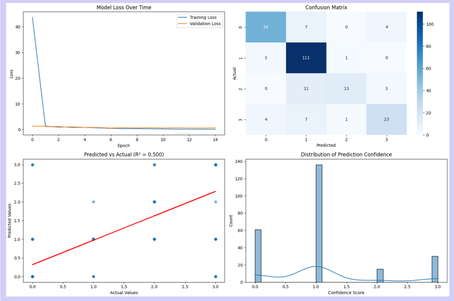


Figure 14 – Training and Validation Loss Graphs, Created by Finalist, 2025

Figure 14 displays training and validation loss over 15 epochs for a model with an 83% accuracy rate, a confusion matrix, prediction accuracy, and confidence distribution. In *Model Loss Over Time* (top-left), the losses converge, showing no severe overfitting. In *Confusion Matrix* (top-right), the diagonal values (54, 111, 13, 23) indicate correct classifications, while off-diagonal values highlight misclassifications. In the *Predicted vs. Actual g*raph, the red line represents an ideal correlation, and the spread of points suggests moderate accuracy (R2=0.5R^2 = 0.5R2=0.5), meaning the model explains 50% of the variance. Peaks around 0, 1, and higher values indicate varying confidence levels, while the density line provides a smooth estimate of prediction confidence distribution.

RESULTS

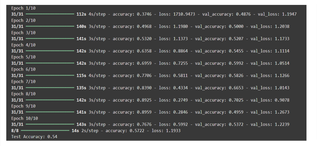


Figure 15 – Model Training 1, Created by Finalist, 2025

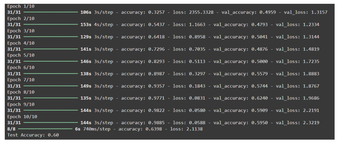


Figure 16 – Model Training 2, Created by Finalist, 2025

Model was trained 4 times. Two trainings for the model with 10 epochs and two trainings with 15 epochs

Highest accuracy percentage was 79% with 15 epochs

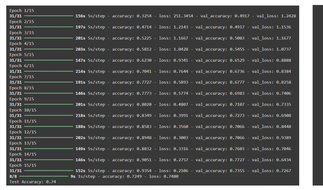


Figure 17 – Model Training 3, Created by Finalist, 2025

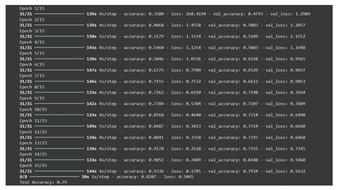


Figure 18 – Model Training 1II , Created by Finalist, 2025

CONCLUSION

- Dataset had a high frequency of neutral images
- Highest accuracy was 79% with 15 epochs of training
- Training and Validation Loss throughout training was high

FUTURE WORK & APPLICATIONS

FUTURE WORK

Pupil Dilation:
- Evaluating tell signs of eyes/pupil dilation

Voice Recognition:
- Emotional cues conveyed through prosody
- Relative pitch, duration, and intensity of speech signal

Other Ways to Evaluate Mental Health:
- Testing blood pressure to depict stress levels

Improving Overall Model Accuracy:
- Using different neural networks to improve model accuracy (such as VGG16)

APPLICATIONS

- Helps clinicians monitor emotional changes in patients with mood disorders
- Helps with remote therapy/telehealth (facial expressions are used to tell a patient's emotional state)
- Serves as a screening tool to detect early signs of mental health conditions such as depression PTSD, and anxiety
- Supports patients who have a hard time communicating (speech-impediment, autism, selective-mutism, and special education students)
- Can supports Alzheimer's and dementia patients to improve care and treatment

REFERENCES

[1] Logo Created by Finalist, 2025

[2] "Long-Term Spaceflight Composite Stress," 2023. Nature.
https://www.nature.com/articles/s41398-023-02638-5

[3] "Convolutional Neural Network (CNN): A Complete Guide," 2023. LearnOpenCV.
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[4] "Measuring Depression Severity," 2022. Frontiers.
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